# Yu Gao

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yu.gao@whoi.edu — (858) 305-1866 — yugaophd.github.io — github.com/yugaophd

## Education

**Ph.D. in Meteorology and Physical Oceanography** — Aug 2016 – Jan 2022
University of Miami — Miami, FL

**Bachelor's Degree in Marine Science** — Aug 2012 – June 2016
Ocean University of China — Qingdao, China

## Work Experience

**Research Associate II-III, The Woods Hole Oceanographic Institution (WHOI)**
March 2024 – Present

- Develop reproducible algorithms for geospatial and oceanographic data across multiple projects.
- Quality control and process buoy, wave glider and mooring observations into high-quality data products.
- Standardize workflows to generate open-access data products for collaborative research.
- Build Python tools for sensor analysis in lab and field settings.
- Developed an AI-powered flood monitoring system integrating real-time news sources and NOAA alerts with OpenAI tools

**Postdoctoral Scholar, UC San Diego** — February 2022 – January 2024

- Developed Python-based data assimilation methods to improve SWOT sea-surface height accuracy.
- Advanced techniques to jointly optimize model fields and correlated error structures in ocean state estimates.

**Research Assistant, Univ. of Miami** — August 2016 – January 2022

- Modeled mesoscale air–sea interactions with a regional coupled ocean–atmosphere model to advance understanding of coupled climate dynamics.
- Quantified mesoscale mixed-layer heat budgets and air–sea heat exchange in the Southern Ocean using coupled climate model output.
- Analyzed mixed-layer depth variability in both regional and climate simulations, highlighting the influence of ocean eddies and atmospheric forcing.

## Peer-reviewed Publications

**Deep Ocean Temperature from the Stratus Ocean Reference Station in the Southeast Pacific cold tongue region (85°W, 20°S), 2012–2025.**, Yu Gao, Albert Plueddemann, Robert Weller, Sebastien Bigorre (in preparation). To be submitted to *Nature Scientific Data.*

**SWOT Data Assimilation with Correlated Error Reduction: Fitting Model and Error Together**, Sarah T. Gille, Yu Gao, Bruce D. Cornuelle, Matthew R. Mazloff, *Journal of Atmospheric and Oceanic Technology 42.3 (2025): 253-268.* DOI:10.1175/JTECH-D-24-0062.1

**Oceanic Mesoscale and Atmospheric Noise Coupling Dampens Southern Ocean Mixed Layer Variability**, Yu Gao, Igor Kamenkovich, Benjamin Kirtman, *Journal of Geophysical Research: Oceans (pre-print)*, DOI:10.22541/essoar.170067051.16403665/v1

**Origins of Mesoscale Mixed-layer Depth Variability in the Southern Ocean**, Yu Gao, Igor Kamenkovich, Natalie Perlin, *Ocean Science, 19, 615-627, 2023.* DOI: 10.5194/os-19-615-2023

**Oceanic Advection Controls Mesoscale Mixed Layer Heat Budget and Air–Sea Heat Exchange in the Southern Ocean**, Yu Gao, Igor Kamenkovich, Natalie Perlin Benjamin Kirtman, *Journal of Physical Oceanography, 52(4), 537-555, 2022a.* DOI: 10.1175/JPO-D-21-0063.1

**A study of mesoscale air–sea interaction in the Southern Ocean with a regional coupled model** Natalie Perlin, Igor Kamenkovich, Yu Gao, Benjamin Kirtman, *Ocean Modelling 153, 101660, 2020.* DOI: 10.1016/j.ocemod.2020.101660

## Data Publications

**Deep-Ocean Temperature data from a mooring in the Southeast Pacific (85°W, 20°S), 2012 - 2025**, Yu Gao, Albert Plueddemann, Robert Weller, Sebastien Bigorre *MBLWHOI Library Dataverse, 2025.* DOI: 10.26027/DATAKX5CQK

**Origins of Mixed Layer Depth Variability in the Southern Ocean**, Yu Gao, Igor Kamenkovich, Benjamin Kirtman, *University of Miami Libraries, 2022b.* DOI: 10.17604/0BKF-P943

**Oceanic Advection Controls Mesoscale Mixed Layer Heat Budget and Air-sea Heat Exchange in the Southern Ocean**
Yu Gao, Igor Kamenkovich, Natalie Perlin, Benjamin Kirtman,
*University of Miami Libraries, 2021.* DOI: 10.17604/94qh-6m66

## Teaching Experience

**MSC 302 Physical Oceanography Laboratory**: I guided and supervised laboratory experiments, and assessed student lab reports and with a focus on enhancing understanding and application of physical oceanography concepts.
Teaching Assistant, University of Miami — Spring 2019

**MSC/ATM 220 Climate and Global Change**: Undergraduate level class on Earth's climate system and the role of natural and anthropogenic processes in shaping climate change. I gave lecture on global climate change, assisted with course materials, and graded assignments.
Teaching Assistant, University of Miami — Fall 2019

## Seminar and Talks

**Mesoscale air-sea Interaction and Mixed Layer Variability in the Southern Ocean**, JPL Center for Climate Sciences Seminar, Pasadena, CA — October 2023

**SWOT Data Assimilation With Correlated Error Reduction**,
NASA-MPOWIR Speaker Series, JPL, Pasadena, CA — November 2022

**Origins of Mesoscale Mixed Layer Variability in the Southern Ocean**,
Ocean Sciences Meeting 2022, Online — Feb-Mar 2022

**SWOT Data Assimilation With Correlated Error Reduction**,
SWOT Science Team Meeting, Chapel Hill, NC, USA — Jun 2022

**Role of Mesoscale Currents in Ocean Mixed Layer Heat Budget**, Ocean Sciences Meeting 2020, San Diego, CA, USA February 2020

## Poster Presentations

**SWOT Data Assimilation with Correlated Error Reduction: Fitting Model and Error Together**, SWOT Science Team Meeting, Toulouse, France Sept 2023

**Origins of Mesoscale Mixed Layer Variability in the Southern Ocean**
US CLIVAR Workshop, Denver, CO, USA Mar 2023

**SWOT Data Assimilation With Correlated Error Reduction**
AGU Fall Meeting, Chicago, IL, USA, Dec 2022

**Role of Mesoscale Currents in Ocean Mixed Layer Heat Budget and Air-Sea Coupling**
AGU Fall Meeting, Online Dec 2020

## Professional Development

**NASA's Earth Observations Summer School, Using Satellite Observations to Advance Climate Models**, Pasadena, CA, USA Aug 16, 17 and 21 - 25, 2023

**The Pattullo Conference by MPOWIR,** Warrenton, VA, USA Sept. 24 - 27, 2023

**Unifying Innovations in Forecasting Capabilities Workshop**
Boulder, CO, USA July 24, 2023 - July 28, 2023

**San Diego Supercomputer Center, Summer Institute 2022, Super-computing and Data Science**, San Diego, CA, August 5 - 9, 2022

**AMS Short Courses: Machine Learning in Python for Environmental Science; Python for Climate and Meteorology** Mar–Apr 2021

**Annual RSMAS Writing Workshop with Dallas Murphy**
Miami, FL, USA and Virtual Dec 2020 - Jan 2021

## Skills

**Programming Languages**: Python, MATLAB, Fortran, LaTeX
**Softwares and computing**: Git, High-performance Computing(HPC)
**Models and Methods**: Regional Ocean Modeling System (ROMS), Finite Volume Community Ocean Model (FVCOM) and data assimilation

## Professional Services

**Referee for:** National Science Foundation & Ocean Science (eISSN: OS 1812-0792, OSD 1812-0822)